

Genome-wide Genotypic Responses to Starvation Selection in *D. melanogaster*

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Introduction

Drosophila melanogaster is a powerful model for studying evolutionary genomics, and Evolve-and-Resequence experiments enable researchers to track genome-wide allele frequency changes under controlled selection pressures. In Hardy et al. (2018), replicate populations were subjected to more than 80 generations of starvation selection, while matched control lines were maintained on abundant food. This strong selection regime produced substantial physiological changes and widespread genomic shifts. In this project, I reanalyzed the pooled whole-genome sequencing data from the starvation-selected and fed control populations to characterize genome-wide patterns of genotypic differentiation, identify regions of elevated divergence, and evaluate functional and evolutionary signatures of adaptation to chronic starvation.

Research Questions

1. Do starvation-selected populations show genome-wide genotypic differentiation from fed control populations?
2. Which genomic regions exhibit the strongest allele frequency divergence between fed and starved populations?
3. Do sliding-window patterns reveal broad selective sweeps or localized genomic regions under strong selection?
4. Which genomic windows show significant enrichment of high-FST SNPs?
5. Are certain variant functional categories enriched for high-FST SNPs?

Methods

Experimental Methods (Hardy et al. 2018)

Flies originated from a diverse outbred population collected in New Jersey. Replicate populations were maintained under two treatments for over 80 generations:

- Starvation Selection: Adults were placed on agar-only media each generation until ~80–90% mortality, and surviving individuals founded the next generation.
- Fed Control: Matched populations maintained on standard food without starvation.

At the end of the experiment, pooled samples of ~100 females per population were sequenced using Illumina paired-end sequencing, generating genome-wide allele frequency data for both starved and fed lines.

Analytical Methods

- Calculated per-SNP FST to measure genomic differentiation between starvation-selected and fed populations.
- Identified highly divergent SNPs using the top one percent of the FST distribution.
- Performed sliding-window FST analysis using 500 kb windows and 100 kb step sizes to detect genomic regions with elevated divergence.
- Measured heterozygosity for each SNP to evaluate loss of genetic diversity.
- Quantified parallel evolution by correlating windowed FST profiles across replicate starvation-selected populations.
- All analyses were run on the Rivanna HPC cluster using SLURM job scripts, R-based data processing and visualization tools

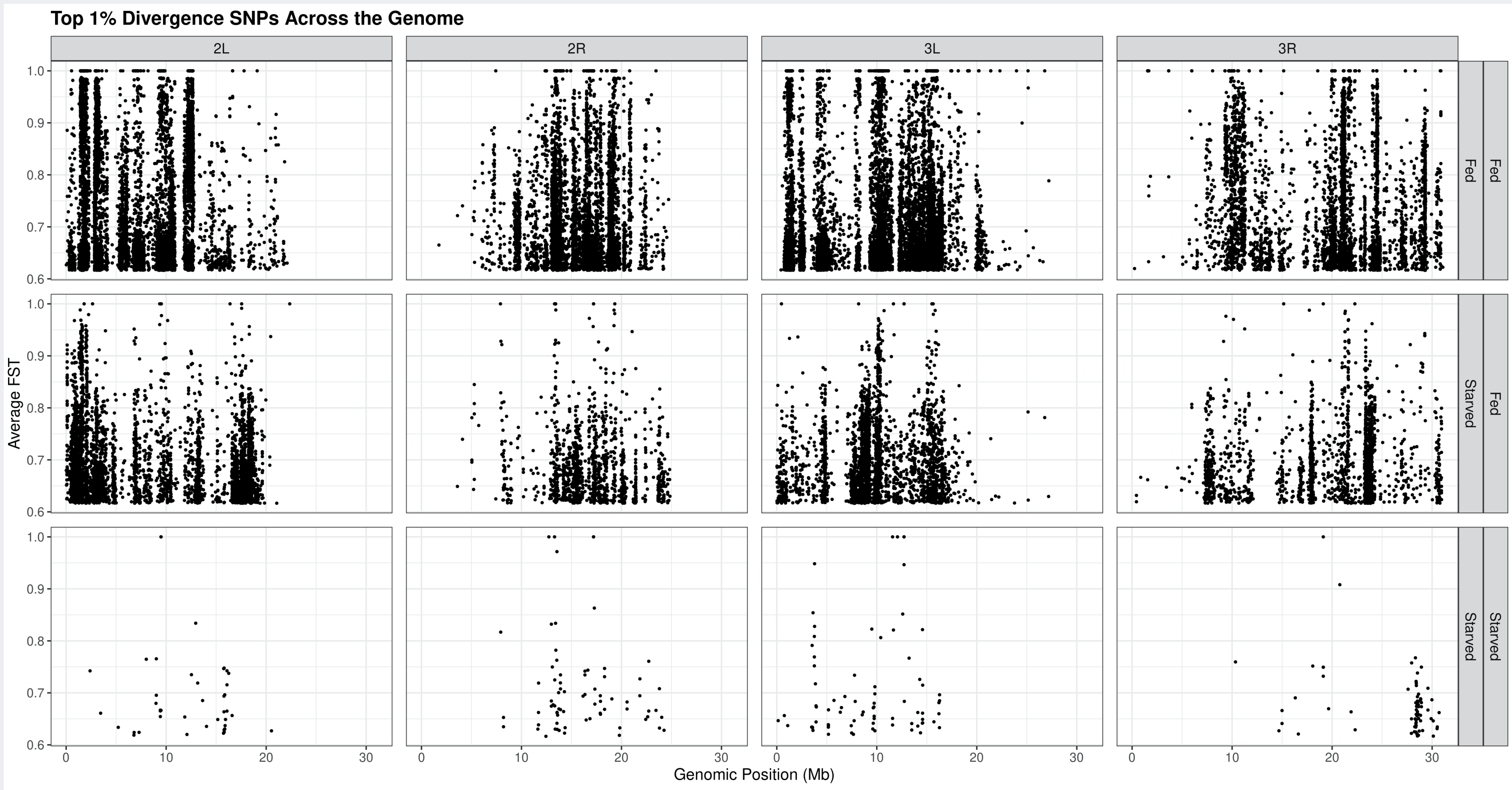


Figure 3. Genome-wide distribution of top 1% high-FST SNPs in *D. melanogaster*. SNPs exceeding the 99th percentile of the genome-wide FST distribution were identified as candidate divergent loci and plotted across chromosome arms for each pairwise population comparison. Distinct clusters of outlier SNPs occur on multiple chromosome arms, indicating localized regions of elevated divergence consistent with selection or linked selective sweeps. Both within-treatment (Fed vs Fed; Starved vs Starved) and between-treatment (Fed vs Starved) contrasts exhibit genomic islands of high divergence, reflecting independent evolutionary trajectories among replicate selection lines as well as treatment-associated differentiation.

Conclusion

PCA analysis revealed clear and consistent separation between fed and starved populations, indicating strong genome-wide differentiation under chronic starvation selection. Divergence was concentrated in specific genomic hotspots: the top 1% FST SNPs clustered on chromosome arms 2L, 3L, and 3R, identifying key regions under intense selection. Sliding-window FST profiles showed narrow, high-magnitude peaks rather than broad sweeps, confirming that responses to starvation are sharply localized.

Window-level enrichment tests revealed several genomic blocks containing significantly more high-FST SNPs than expected by chance, further highlighting discrete targets of selection. Functional analyses showed that missense, splice-site, and start/stop codon variants exhibit higher FST values than synonymous and intergenic SNPs, suggesting that starvation selection disproportionately affects coding and regulatory sites.

Results

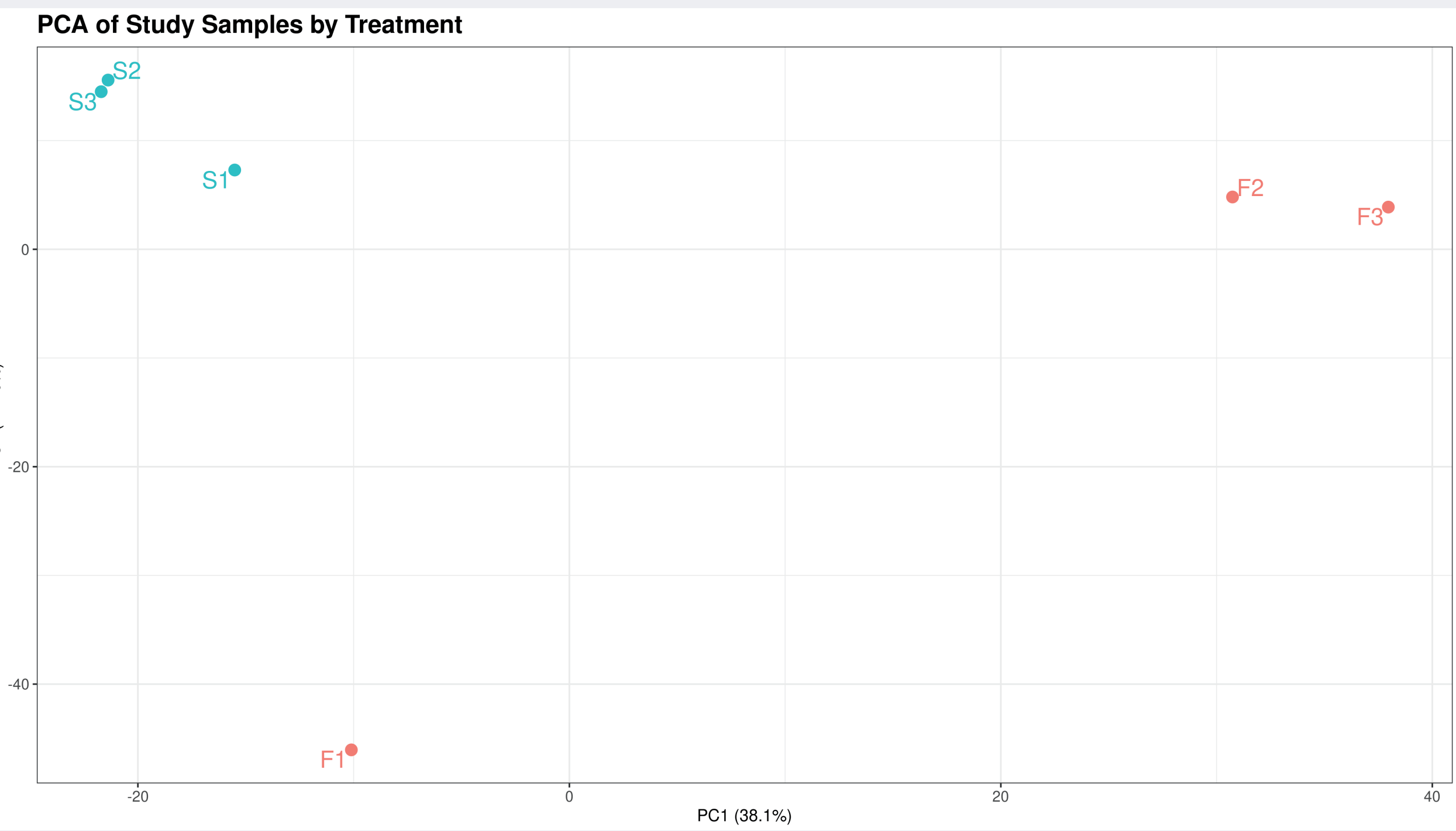


Figure 1. Genome-wide differentiation between starvation-selected and fed populations PCA of genome-wide allele frequencies shows clear separation between starvation-selected (S) and fed control flies (F) along the major axes of variation, indicating strong genome-wide genotypic differentiation after >80 generations of divergent selection.

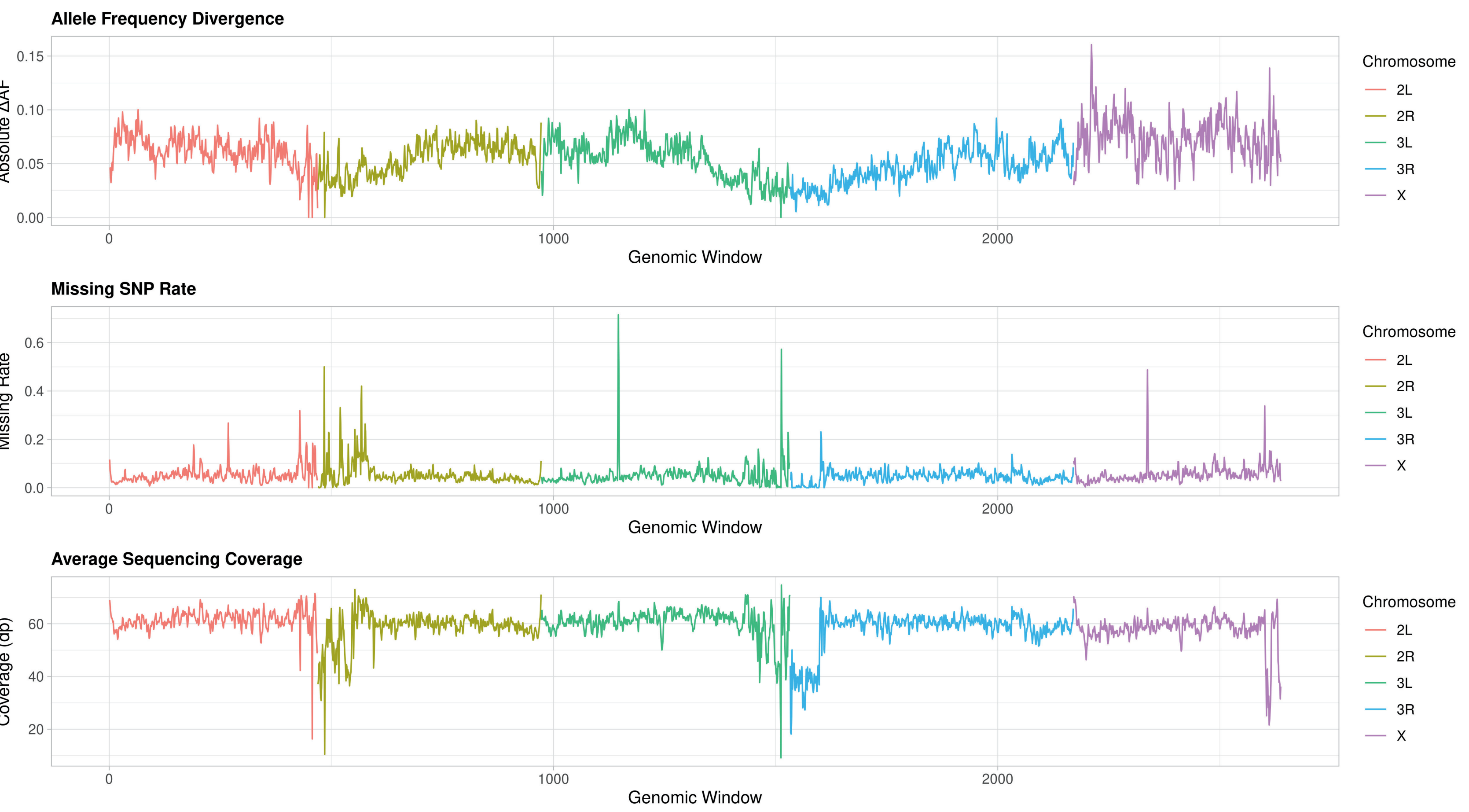


Figure 2. Sliding-window allele frequency change and sequencing quality across the genome The top panel shows the average absolute allele frequency change between starvation-selected and fed control populations across 500-kilobase windows. Divergence varies strongly across the genome, with several localized peaks, especially on chromosome X, indicating region-specific responses to starvation selection rather than broad genome-wide sweeps.

The middle and bottom panels show missing-site rates and sequencing coverage for each window. Peaks in missing data and drops in coverage occur in a few narrow regions and help confirm that major divergence peaks in the top panel are biological rather than artifacts of sequencing quality.

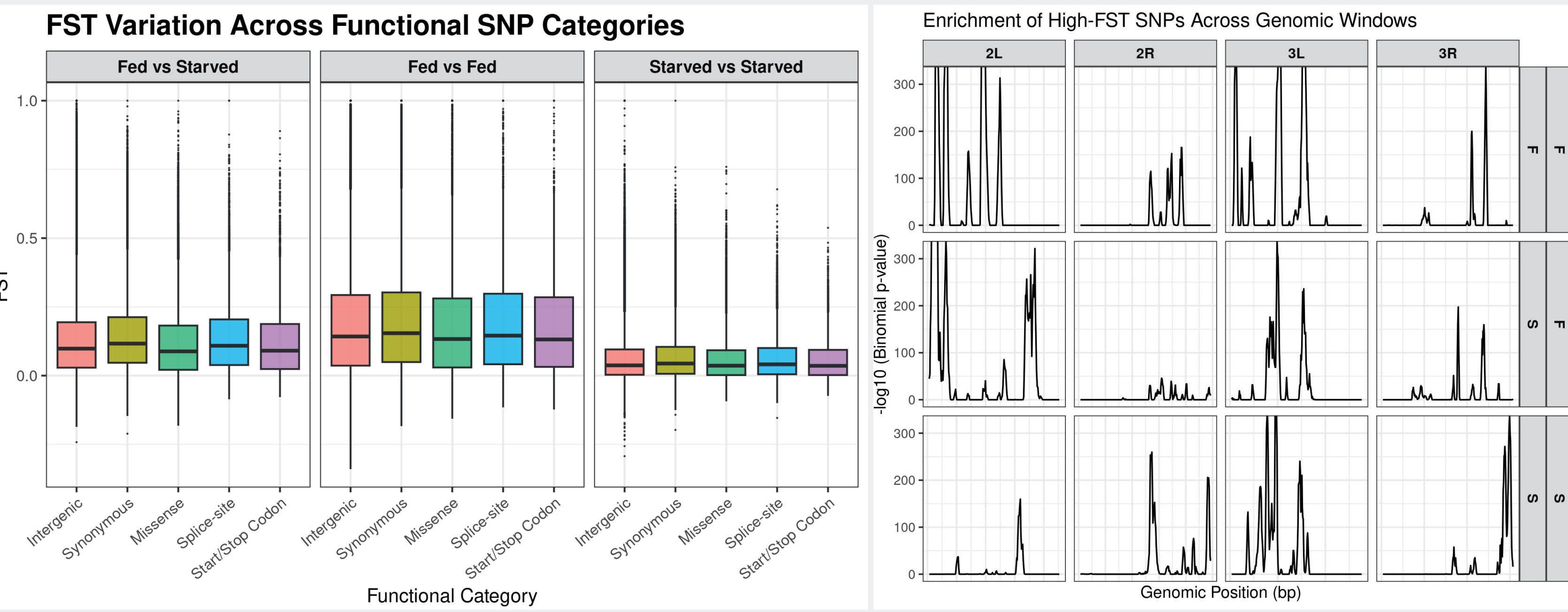


Figure 4. Genome-wide FST variation across major functional SNP classes in fed and starved selection lines of *D. melanogaster*. FST distributions were broadly similar among functional categories, with intergenic and synonymous variants showing the lowest divergence and missense, splice-site, and start/stop codon variants showing slightly higher median FST.

The three comparisons all show appreciable differentiation because each contrast reflects independently evolving experimental populations rather than technical replicates. Thus, both within-treatment and between-treatment comparisons capture genuine evolutionary divergence accumulated during experimental evolution.

Figure 5. Window-based enrichment of high-FST SNPs across the *D. melanogaster* genome. A sliding 500 kb window was used to evaluate enrichment of high-FST SNPs relative to neutral expectations. A binomial test was applied to each window, and $-\log_{10}(p)$ values indicate windows containing significantly more high-FST SNPs than expected by chance. Sharp peaks highlight genomic regions under strong localized selection.

References

Hardy, C. M., Burke, M. K., Everett, L. J., Han, M. V., Lantz, K. M., & Gibbs, A. G. (2018). Genome-Wide Analysis of Starvation-Selected *Drosophila melanogaster*—A Genetic Model of Obesity. *Molecular biology and evolution*, 35(1), 50–65. <https://doi.org/10.1093/molbev/msx254>

Acknowledgments

Thank you to Dr. Bergland and Robert Porter for their guidance and support throughout this project.